Chapter 1. Multivariate Data and the Geometry of the Sample

Data matrices, descriptive statistics, projections, sphering, Mahalanobis distance, eigendecomposition

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1 Multivariate Data and the Goals of Analysis

Multivariate (multidimensional) data are measurements of two or more variables on each individual or experimental unit. The goals of analysis fall into two groups:

(i) **Inference about an unknown population**: understand and infer features of the population from which the data were drawn.

(ii) **Prediction of future observations**: predict the value or category of new observations.

In either case one starts with exploratory data analysis (EDA) to find structure, outliers and relationships among variables. Numerical summaries are descriptive statistics (mean vector, covariance matrix, correlation matrix); graphical summaries are scatterplot matrices, principal component plots, heatmaps, correlation plots and trellis graphs.

Example 1.1: two running examples (slides Examples 1.1, 1.2)

Penguins data. For $n = 333$ penguins from the Palmer archipelago there are 4 numerical variables (bill length, bill depth, flipper length, body mass), 3 categorical variables (species, island, sex) and year. Only the numerical variables go into the data matrix $\mathbf{X}$; categorical variables are kept as separate vectors. This is the typical case $n > p$.
Golub gene expression data (1999). For 72 leukemia patients the expression levels of 7129 genes were measured, together with a categorical label ALL/AML. This is high-dimensional, low-sample-size data ($p = 7129 \gg n = 72$). Scientific question: can the leukemia type be predicted from gene expression alone?

2 The Data Matrix and Descriptive Statistics

Motivation. Multivariate data are an $n \times p$ matrix, and almost every method in the course is a statement about this matrix. Writing the sample mean and covariance as matrix products lets us use linear algebra instead of indices.

2.1 Data matrix

Measurements of p variables on n items form an $n \times p$ matrix

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} = [\mathbf{x}_{(1)}, \ldots, \mathbf{x}_{(p)}].$$

The ith row $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top \in \mathbb{R}^p$ is the **observation vector** of item i; the jth column $\mathbf{x}_{(j)} = (x_{1j}, \ldots, x_{nj})^\top \in \mathbb{R}^n$ is the **variable vector** of variable j. All variables in $\mathbf{X}$ are taken to be numerical.

2.2 Sample mean, covariance and correlation

Definition 1.2: descriptive statistics

$$\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i, \qquad \mathbf{S} = (s_{jk}), \quad s_{jk} = \frac{1}{n-1}\sum_{i=1}^{n}(x_{ij}-\bar{x}_j)(x_{ik}-\bar{x}_k),$$

$$\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i-\bar{\mathbf{x}})(\mathbf{x}_i-\bar{\mathbf{x}})^\top, \qquad \mathbf{R} = (r_{jk}), \quad r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}}\sqrt{s_{kk}}}, \qquad \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2},$$

with $\mathbf{D} = \operatorname{diag}(s_{11},\dots,s_{pp})$. $\mathbf{R}$ is defined only when all $s_{jj} > 0$. One also uses $\mathbf{S}_n = \frac{n-1}{n}\mathbf{S}$ (divisor n).

Remark 1.3: a Gaussian with covariance S always exists

$\mathbf{S}$ is symmetric and positive semidefinite, so a symmetric square root $\mathbf{S}^{1/2}$ exists (§5). With $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\mathbf{X} = \boldsymbol{\mu} + \mathbf{S}^{1/2}\mathbf{Z}$, $\mathrm{E}\mathbf{X} = \boldsymbol{\mu}$ and $\mathrm{Cov}\mathbf{X} = \mathbf{S}^{1/2}\mathbf{I}_p\mathbf{S}^{1/2} = \mathbf{S}$.

2.3 The centering matrix

Let $\mathbf{1}_n = (1,\dots,1)^\top$, $\mathbf{J}_n = \mathbf{1}_n\mathbf{1}_n^\top$ and $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{J}_n$.

Theorem 1.4: matrix forms of $\bar{\mathbf{x}}$ and S (slides Exercise 1.1)

$$\bar{\mathbf{x}} = \tfrac{1}{n}\mathbf{X}^\top\mathbf{1}_n, \qquad \mathbf{X}_C := \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top = \mathbf{H}_n\mathbf{X}, \qquad \mathbf{S} = \tfrac{1}{n-1}\mathbf{X}^\top\mathbf{H}_n\mathbf{X} = \tfrac{1}{n-1}\mathbf{X}_C^\top\mathbf{X}_C,$$

and $\mathbf{H}_n^\top = \mathbf{H}_n$, $\mathbf{H}_n^2 = \mathbf{H}_n$.

Proof. The jth entry of $\mathbf{X}^\top\mathbf{1}_n$ is $\sum_i x_{ij}$, giving the first identity. Hence $\mathbf{X}_C = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top = \mathbf{X} - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\mathbf{X} = \mathbf{H}_n\mathbf{X}$. Idempotency follows from $\mathbf{1}_n^\top\mathbf{1}_n = n$:

$$\left(\mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n\right)^2 = \mathbf{I}_n - \tfrac{2}{n}\mathbf{J}_n + \tfrac{1}{n^2}\mathbf{1}_n(\mathbf{1}_n^\top\mathbf{1}_n)\mathbf{1}_n^\top = \mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n.$$

Then $\mathbf{X}_C^\top\mathbf{X}_C = \mathbf{X}^\top\mathbf{H}_n^\top\mathbf{H}_n\mathbf{X} = \mathbf{X}^\top\mathbf{H}_n\mathbf{X}$, and $\mathbf{X}_C^\top\mathbf{X}_C = \sum_i(\mathbf{x}_i-\bar{\mathbf{x}})(\mathbf{x}_i-\bar{\mathbf{x}})^\top$. ■

[Supplement] A symmetric idempotent matrix has eigenvalues 0 or 1, so $\operatorname{rank}(\mathbf{H}_n) = \operatorname{tr}(\mathbf{H}_n) = n-1$.

3 Sample Geometry in p-Space

3.1 The data ellipse

Each $\mathbf{x}_i$ is a point in $\mathbb{R}^p$. The mean vector is the center of gravity of the points and the covariance matrix describes their spread. The **data ellipse**

$$\{\mathbf{x} \in \mathbb{R}^p : (\mathbf{x}-\bar{\mathbf{x}})^\top\mathbf{S}^{-1}(\mathbf{x}-\bar{\mathbf{x}}) = c\}$$

summarizes location, shape and size for fixed $c > 0$ (an ellipsoid for $p \geq 3$). The penguin plot uses $c = 5.991 = \chi^2_2(0.05)$. If the cloud is not elliptical, the mean and covariance alone describe it poorly.

Example 1.5: data ellipses

Draw the ellipses for mean $\mathbf{0}$ and $\mathbf{S}_1 = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathbf{S}_2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$.

[Supplement] For $\mathbf{S}_1$ the axes are the coordinate axes with half-length ratio $3:1$. $\mathbf{S}_2$ has eigenvalues $9, 1$ with eigenvectors $(1,1)^\top/\sqrt{2}$, $(1,-1)^\top/\sqrt{2}$: the same ellipse rotated by $45°$. Half-axis lengths are $\sqrt{c\lambda_j}$.

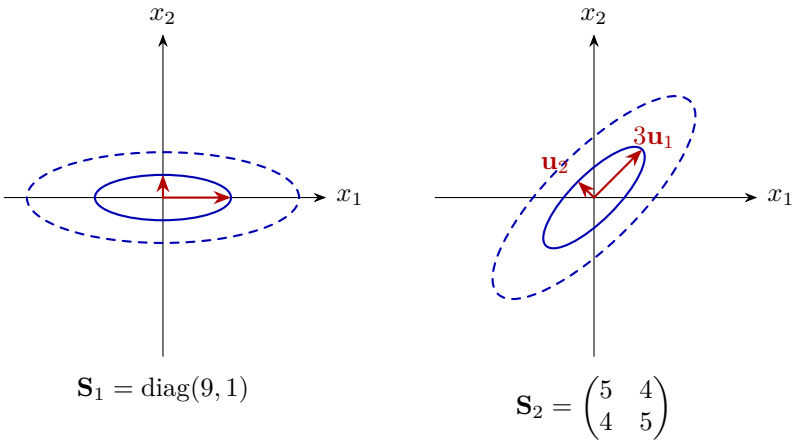


Figure. *Data ellipses for $c = 1$ (solid) and $c = 4$ (dashed). The covariance 4 in $\mathbf{S}_2$ rotates the axes by $45°$.*

3.2 Subspaces, orthogonal projection and projection scores

Definition 1.6: subspace and affine subspace

- **Subspace**: $\mathcal{L}_q = \mathrm{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_q\} = \{\mathbf{Vc} : \mathbf{c} \in \mathbb{R}^q\}$, $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_q]$ (linearly independent). It contains the origin.
- **Affine subspace**: $\mathcal{L}_q(\mathbf{x}) = \mathbf{x} + \mathcal{L}_q$, e.g. a line $\{\mathbf{x} + t\mathbf{v}\}$ or a plane $\{\mathbf{x} + c_1\mathbf{v}_1 + c_2\mathbf{v}_2\}$. It is not a vector space unless it contains the origin, but has the same shape as $\mathcal{L}_q$. Its representation (base point, spanning vectors) is not unique.

Definition 1.7: orthogonal projection and projection score

Projection onto $\mathrm{span}\{\mathbf{v}\}$ and the projection score (signed distance):

$$P_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}\mathbf{v}^\top}{\mathbf{v}^\top\mathbf{v}}\mathbf{x} = s_{\mathbf{v}}(\mathbf{x})\frac{\mathbf{v}}{\sqrt{\mathbf{v}^\top\mathbf{v}}}, \qquad s_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}^\top\mathbf{x}}{\sqrt{\mathbf{v}^\top\mathbf{v}}}.$$

The projection matrix onto $\mathcal{L}_q$ is $\mathbf{P}_{\mathbf{V}} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$; for an orthonormal basis ($\mathbf{V}^\top\mathbf{V} = \mathbf{I}_q$)

$$P_{\mathcal{L}_q}(\mathbf{x}) = \mathbf{V}\mathbf{V}^\top\mathbf{x} = s_{\mathbf{v}_1}(\mathbf{x})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x})\mathbf{v}_q,$$

so $\mathbf{V}^\top\mathbf{x} = (s_{\mathbf{v}_1}(\mathbf{x}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}))^\top$ are the coordinates in the subspace. The projection does not depend on the basis; the scores do. A scatterplot of variables i, j is the projection onto $\mathrm{span}\{\mathbf{e}_i, \mathbf{e}_j\}$.

Derivation: why P_V is the orthogonal projection [Supplement]

$\mathbf{P}_\mathbf{V}^\top = \mathbf{P}_\mathbf{V}$ and $\mathbf{P}_\mathbf{V}^2 = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top\mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top = \mathbf{P}_\mathbf{V}$. Also $\mathbf{P}_\mathbf{V}\mathbf{x} \in \mathcal{L}_q$, and the residual satisfies $\mathbf{V}^\top(\mathbf{x} - \mathbf{P}_\mathbf{V}\mathbf{x}) = \mathbf{0}$, so it is orthogonal to $\mathcal{L}_q$. By Pythagoras, for any $\mathbf{Vc}$, $\|\mathbf{x} - \mathbf{Vc}\|^2 = \|\mathbf{x} - \mathbf{P}_\mathbf{V}\mathbf{x}\|^2 + \|\mathbf{P}_\mathbf{V}\mathbf{x} - \mathbf{Vc}\|^2 \geq \|\mathbf{x} - \mathbf{P}_\mathbf{V}\mathbf{x}\|^2$: $\mathbf{P}_\mathbf{V}\mathbf{x}$ is the closest point of $\mathcal{L}_q$ to $\mathbf{x}$.

3.3 Projection onto an affine subspace and decomposition of observations

Projection onto $\mathcal{L}_q(\bar{\mathbf{x}})$ is done in two steps:

(1) translate $\mathbf{x} \mapsto \mathbf{x} - \bar{\mathbf{x}}$ and project onto $\mathcal{L}_q$;

(2) translate back by $\bar{\mathbf{x}}$:

$$P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{P}_\mathbf{V}(\mathbf{x} - \bar{\mathbf{x}}) = \bar{\mathbf{x}} + s_{\mathbf{v}_1}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_q.$$

Hence an observation splits into projection plus remainder, and by Pythagoras, with $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$,

$$\mathbf{x} = \bar{\mathbf{x}} + \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_j + r_{\mathcal{L}_q}(\mathbf{x}), \qquad d(\mathbf{x}, \bar{\mathbf{x}})^2 = \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})^2 + \|r_{\mathcal{L}_q}(\mathbf{x})\|^2.$$

This decomposition is the starting point of dimension reduction (PCA).

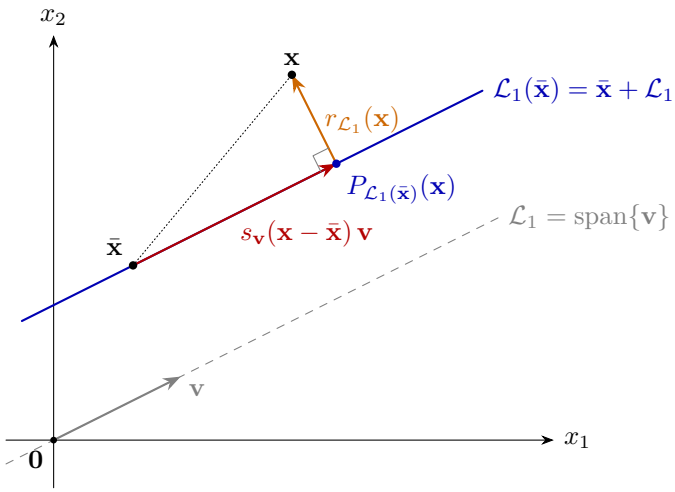


Figure. *Projection onto the affine line $\mathcal{L}_1(\bar{\mathbf{x}})$: the observation splits into $\bar{\mathbf{x}}$, the projected part and an orthogonal remainder.*

Remark 1.8: geometric meaning of variance

Projection scores are tied to conditional variances. If $X_1 \sim N(0, \sigma_1^2)$ and $X_2 = X_1 + \epsilon_2$ with independent $\epsilon_2 \sim N(0, \sigma_2^2)$, then $\mathrm{Var}(X_2 \mid X_1) = \sigma_2^2$ and $\mathrm{Var}(X_1 \mid X_2) = \dfrac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$.

[Supplement] The latter is $\sigma_1^2 - \sigma_1^4/(\sigma_1^2 + \sigma_2^2)$ (conditional distribution formula, Ch. 3): what remains after removing the projected part is the conditional variance.

3.4 Linear and affine transformations

A $q \times p$ matrix $\mathbf{L}$ defines the linear transformation $\mathbf{x} \mapsto \mathbf{Lx}$; $q > p$ only adds dimensions without information and is rarely used. Examples: the projection $\mathbf{x} \mapsto \mathbf{P}_\mathbf{V}\mathbf{x}$ and the score map $\mathbf{x} \mapsto \mathbf{S}_\mathbf{V}\mathbf{x}$,

$\mathbf{S_V} = (\mathbf{v}_1/\|\mathbf{v}_1\|, \ldots, \mathbf{v}_q/\|\mathbf{v}_q\|)^\top$. Projection onto an affine subspace is not linear (because of the translation) but an **affine transformation** $\mathbf{x} \mapsto \boldsymbol{m} + \mathbf{L}\mathbf{x}$.

Theorem 1.9: affine maps and standardization (slides Exercise 1.2)

1. Orthogonal projection onto $\mathcal{L}_q(\bar{\mathbf{x}})$ is affine with $\boldsymbol{m} = \bar{\mathbf{x}} - \mathbf{P_V}\bar{\mathbf{x}}$, $\mathbf{L} = \mathbf{P_V} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$; so is the score map $\mathbf{x} \mapsto (s_{\mathbf{v}_1}(\mathbf{x}-\bar{\mathbf{x}}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}-\bar{\mathbf{x}}))^\top$.
2. (Standardization) The data $\mathbf{x}_i' = \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}})$ have sample mean $\mathbf{0}$ and unit sample variances, and $\mathbf{x}' = \mathbf{D}^{-1/2}\mathbf{x} - \mathbf{D}^{-1/2}\bar{\mathbf{x}}$ is affine.

Proof. (1) $P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{P_V}(\mathbf{x}-\bar{\mathbf{x}}) = (\bar{\mathbf{x}} - \mathbf{P_V}\bar{\mathbf{x}}) + \mathbf{P_V}\mathbf{x}$; the score vector is $\mathbf{S_V}\mathbf{x} - \mathbf{S_V}\bar{\mathbf{x}}$.

(2) $\bar{\mathbf{x}}' = \frac{1}{n}\sum_i \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}}) = \mathbf{0}$; the jth sample variance is $s_{jj}^{-1/2} s_{jj} s_{jj}^{-1/2} = 1$. ■

3.5 Rotation of coordinates

For $p = 2$ and $\mathbf{R}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, put $\mathbf{v}_1 = \mathbf{R}_\theta\mathbf{e}_1$, $\mathbf{v}_2 = \mathbf{R}_\theta\mathbf{e}_2$; then $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2]$ satisfies $\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{I}_2$. Since $\mathbf{x} = \mathbf{V}\mathbf{V}^\top\mathbf{x} = [\mathbf{v}_1, \mathbf{v}_2](\mathbf{v}_1^\top\mathbf{x}, \mathbf{v}_2^\top\mathbf{x})^\top$, the coordinates of $\mathbf{x}$ in the new system are the projection scores $(\mathbf{v}_1^\top\mathbf{x}, \mathbf{v}_2^\top\mathbf{x})$. A linear map by an orthonormal basis is a rotation of coordinates.

4 Standardization, Sphering and Mahalanobis Distance

4.1 Univariate standardization and its limitation

For $N(\mu, \sigma^2)$ the distance of x from the center is better measured by the **standardized distance** $|x-\mu|/\sigma$ than by $|x-\mu|$. In the multivariate case, standardizing each variable, $\mathbf{x} \mapsto \mathbf{D}^{-1/2}(\mathbf{x}-\bar{\mathbf{x}})$, removes scale differences but **ignores correlation**, so it does not give a distance that reflects the distribution.

Example 1.10: distances in bivariate data (slides Example 1.3)

$\bar{\mathbf{x}} = \mathbf{0}$, $\mathbf{S} = \begin{pmatrix} 640000 & 9600 \\ 9600 & 225 \end{pmatrix} = \begin{pmatrix} 800^2 & 800\cdot 15\cdot 0.8 \\ 800\cdot 15\cdot 0.8 & 15^2 \end{pmatrix}$ (correlation 0.8). For $\mathbf{y} = (-500, 10)^\top$, $\mathbf{z} = (1500, 15)^\top$: $d(\bar{\mathbf{x}}, \mathbf{y}) = 500.1$, $d(\bar{\mathbf{x}}, \mathbf{z}) = 1500.075$, so $\mathbf{z}$ looks farther. After standardization $\mathbf{y}' = (-0.625, 0.667)^\top$, $\mathbf{z}' = (1.875, 1.000)^\top$ and still $d(\mathbf{0}, \mathbf{y}') = 0.91 < 2.125 = d(\mathbf{0}, \mathbf{z}')$. Yet the scatterplot has many points near $\mathbf{z}$ and almost none near $\mathbf{y}$ ($\mathbf{y}$ goes against the positive correlation). After standardization the SD in the direction of $\mathbf{y}'$ is 0.36 and in the direction of $\mathbf{z}'$ is 1.31, not 1: standardization does not standardize spread in every direction. After sphering (Example 1.14), $\|\mathbf{y}' - \bar{\mathbf{x}}'\| = 2.65 > 2.18 = \|\mathbf{z}' - \bar{\mathbf{x}}'\|$: now $\mathbf{y}$ is correctly farther. [Supplement] These numbers use the covariance of the plotted sample; the rounded $\mathbf{S}$ above gives 2.04 and 2.05.

4.2 Sphering

If the covariance is $\mathbf{I}_p$, the data ellipse is a sphere ($c = \|\mathbf{x}-\bar{\mathbf{x}}\|^2$) and spread is 1 in **every** direction.

Theorem 1.11: isotropy of sphered data (slides homework)

If $\bar{\mathbf{x}} = \mathbf{0}$ and $\mathbf{S} = \mathbf{I}_p$, then for any unit vector $\mathbf{v}$ the scores $S_{\mathbf{v}}(\mathbf{x}_i) = \mathbf{v}^\top \mathbf{x}_i$ have sample variance 1.

Proof. The scores have mean $\mathbf{v}^\top \bar{\mathbf{x}} = 0$ and sample variance $\mathbf{v}^\top \mathbf{S} \mathbf{v} = \mathbf{v}^\top \mathbf{v} = 1$. ■

Example 1.12: sphering a diagonal covariance (slides Example 1.5)

$\bar{\mathbf{x}} = (10, 10)^\top$, $\mathbf{S} = \operatorname{diag}(4, 9)$: the sphering matrix is $\mathbf{D}^{-1/2} = \operatorname{diag}(1/2, 1/3)$ and $\mathbf{S}' = \mathbf{D}^{-1/2} \mathbf{S} \mathbf{D}^{-1/2} = \mathbf{I}_2$.

Definition 1.13: sphering transformation

A transformation after which the covariance is $\mathbf{I}_p$. For general $\mathbf{S}$ it is built in three steps.

1. **Rotation**: choose orthogonal $\mathbf{U}_S = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ so that $\mathbf{x}_i' = \mathbf{U}_S^\top \mathbf{x}_i$ has diagonal sample covariance $\mathbf{D}_S = \mathbf{U}_S^\top \mathbf{S} \mathbf{U}_S = \operatorname{diag}(d_{11}, \ldots, d_{pp})$ (the eigendecomposition of $\mathbf{S}$).
2. **Standardization**: $\mathbf{x}' \mapsto \mathbf{D}_S^{-1/2} \mathbf{x}'$.
3. **Back-rotation**: $\mathbf{x} \mapsto \mathbf{U}_S \mathbf{x}$.

Altogether $\mathbf{x} \mapsto \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \mathbf{x} = \mathbf{S}^{-1/2} \mathbf{x}$.

[Supplement] The sphered covariance is $\mathbf{I}_p$.

$$\begin{aligned} \mathbf{S}' &= \tfrac{1}{n-1} \textstyle\sum_i \mathbf{S}^{-1/2} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1/2} = \mathbf{S}^{-1/2} \mathbf{S} \mathbf{S}^{-1/2} \\ &= \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top = \mathbf{U}_S \mathbf{U}_S^\top = \mathbf{I}_p. \end{aligned}$$

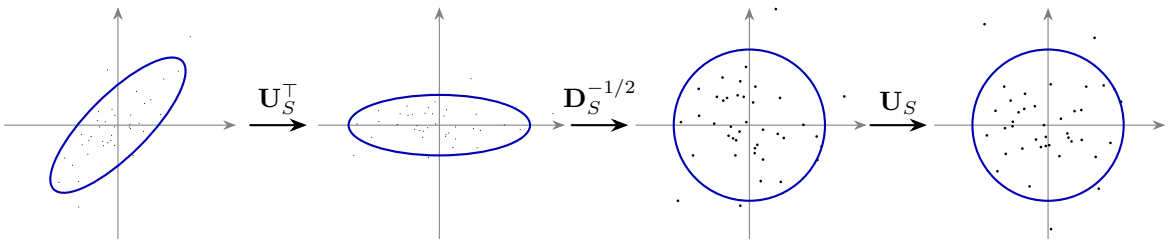
■

Example 1.14: applying the sphering transformation (slides Example 1.6)

$\bar{\mathbf{x}} = \mathbf{0}$, $\mathbf{S} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. With $\mathbf{U}_S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ (rotation by $\theta = \pi/4$),

$$\mathbf{U}_S^\top \mathbf{S} \mathbf{U}_S = \tfrac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{D}_S,$$

so sphering is $\mathbf{x} \mapsto \mathbf{U}_S \operatorname{diag}(1/3, 1) \mathbf{U}_S^\top \mathbf{x}$.



(a) data, cov $\mathbf{S}$ (b) rotated, cov $\operatorname{diag}(9, 1)$ (c) standardized, cov $\mathbf{I}_2$ (d) rotated back, cov $\mathbf{I}_2$

Figure. *The three steps of sphering for $\mathbf{S}$ of Example 1.14 (40 points with this sample covariance). The overall map $\mathbf{S}^{-1/2}$ turns the ellipse into a circle.*

4.3 Mahalanobis distance

Definition 1.15: Mahalanobis (statistical) distance

For a distribution with mean $\bar{\mathbf{x}}$ and covariance $\mathbf{S}$, the Mahalanobis distance of $\mathbf{y}$ from $\bar{\mathbf{x}}$ is the Euclidean distance after sphering:

$$d_S(\mathbf{y}, \bar{\mathbf{x}}) = \|\mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top (\mathbf{y} - \bar{\mathbf{x}})\|, \qquad d_S(\mathbf{y}, \bar{\mathbf{x}})^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{y} - \bar{\mathbf{x}}).$$

Proof. $d_S^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top \mathbf{U}_S \mathbf{D}_S^{-1/2} \underbrace{\mathbf{U}_S^\top \mathbf{U}_S}_{\mathbf{I}} \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top (\mathbf{y} - \bar{\mathbf{x}})$ and $\mathbf{U}_S \mathbf{D}_S^{-1} \mathbf{U}_S^\top = \mathbf{S}^{-1}$. ■

Sphering and Mahalanobis distance are the multivariate counterparts of studentisation and standardized distance.

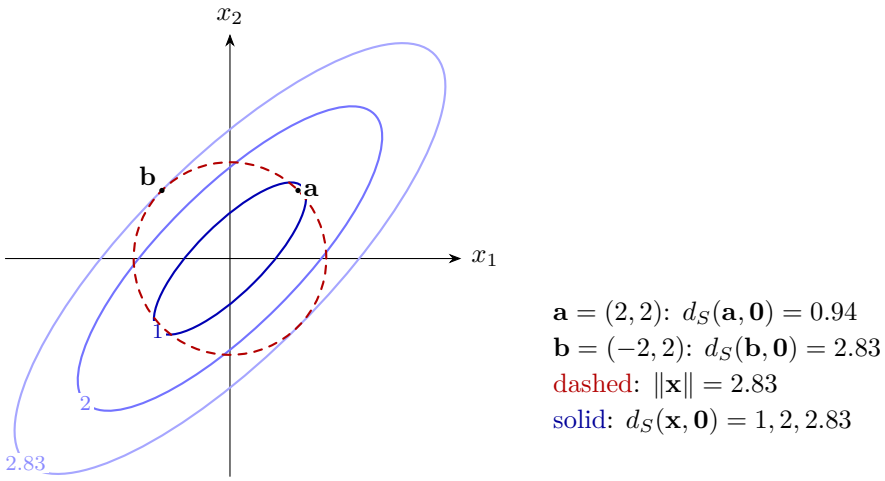


Figure. *Euclidean versus Mahalanobis distance for* $\mathbf{S} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$*:* $\mathbf{a}$ *and* $\mathbf{b}$ *are equally far in Euclidean distance, but* $\mathbf{b}$ *lies against the correlation.*

5 Eigendecomposition and Positive Definiteness

Motivation. A covariance matrix is symmetric and positive semidefinite, so it has an orthonormal eigenbasis. The eigenvectors are the natural axes of the data cloud and the eigenvalues the variances along them; PCA, Mahalanobis distance and the normal density all rest on this decomposition.

5.1 Eigendecomposition

For symmetric $\mathbf{A}$, $\mathbf{Au} = \lambda\mathbf{u}$ ($\mathbf{u} \neq \mathbf{0}$) makes $\mathbf{u}$ an eigenvector and λ an eigenvalue: $\mathbf{A}$ only stretches or shrinks $\mathbf{u}$, by the factor λ.

Theorem 1.16: eigendecomposition of a symmetric matrix (slides homework)

A symmetric $\mathbf{A} \in \mathbb{R}^{p \times p}$ satisfies $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top = \sum_{i=1}^{p} \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$ with $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ orthogonal and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$ real.

- If $\mathbf{u}_i$ is an eigenvector so is $c\mathbf{u}_i$; normalized to length 1 it is unique up to sign (for distinct eigenvalues).
- Eigenvectors of distinct eigenvalues are orthogonal.

- With repeated eigenvalues the decomposition exists but $\mathbf{U}$ is not unique, e.g. $\sigma^2\mathbf{I}_p = \mathbf{U}(\sigma^2\mathbf{I})\mathbf{U}^\top$ for every orthogonal $\mathbf{U}$.

Orthogonality. From $\mathbf{A}\mathbf{u}_j = \lambda_j\mathbf{u}_j$, $\mathbf{u}_i^\top\mathbf{A}\mathbf{u}_j = \lambda_j\mathbf{u}_i^\top\mathbf{u}_j$; from $\mathbf{A}^\top = \mathbf{A}$, $\mathbf{u}_i^\top\mathbf{A} = \lambda_i\mathbf{u}_i^\top$, so $\mathbf{u}_i^\top\mathbf{A}\mathbf{u}_j = \lambda_i\mathbf{u}_i^\top\mathbf{u}_j$. Subtracting, $(\lambda_i - \lambda_j)\mathbf{u}_i^\top\mathbf{u}_j = 0$. ■

Example 1.17: eigenstructure of the sphering matrix (slides Example 1.7)

$\mathbf{A} = \mathbf{S}^{-1/2} = \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix}$ (Example 1.14), $\mathbf{u}_1 = (1,1)^\top/\sqrt{2}$: $\mathbf{A}\mathbf{u}_1 = \frac{1}{\sqrt{2}}(2/3-1/3,\ -1/3+2/3)^\top = \frac{1}{3}\mathbf{u}_1$.
[Supplement] Similarly $\mathbf{A}\mathbf{u}_2 = \mathbf{u}_2$. Sphering rescales each eigen-direction by $1/\sqrt{\lambda_j}$.

5.2 Inverse, square root, definiteness

If no eigenvalue is 0, $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}^{-1}\mathbf{U}^\top$.

Definition 1.18: NND and PD

A symmetric $\mathbf{A}$ is **non-negative definite (NND)** if $\mathbf{x}^\top\mathbf{A}\mathbf{x} \geq 0$ for all $\mathbf{x}$, and **positive definite (PD)** if $\mathbf{x}^\top\mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Theorem 1.19: eigenvalues and definiteness (slides homework)

A symmetric $\mathbf{A}$ with all eigenvalues positive is PD; with all eigenvalues non-negative it is NND.

Proof. With $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ and $\mathbf{y} = \mathbf{U}^\top\mathbf{x}$ (so $\mathbf{y} = \mathbf{0} \iff \mathbf{x} = \mathbf{0}$), $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \mathbf{y}^\top\mathbf{\Lambda}\mathbf{y} = \sum_i \lambda_i y_i^2$. If all $\lambda_i > 0$ and $\mathbf{x} \neq \mathbf{0}$ some $y_i \neq 0$ and the sum is positive; NND is the same. ■

[Supplement] Conversely, if $\mathbf{A}$ is PD (NND), $\lambda_i = \mathbf{u}_i^\top\mathbf{A}\mathbf{u}_i > 0$ (≥ 0): the conditions are equivalent.

Theorem 1.20: the sample covariance is always NND, not necessarily PD (slides homework)

Proof. $\mathbf{z}^\top\mathbf{S}\mathbf{z} = \frac{1}{n-1}\sum_i\{(\mathbf{x}_i - \bar{\mathbf{x}})^\top\mathbf{z}\}^2 \geq 0$. But if the data lie in a lower-dimensional subspace of $\mathbb{R}^p$ (e.g. on a line), any $\mathbf{z} \neq \mathbf{0}$ orthogonal to it gives $\mathbf{z}^\top\mathbf{S}\mathbf{z} = 0$. ■

When n is much larger than p, $\mathbf{S}$ is PD in most cases and $\mathbf{S}^{-1}$ exists.

Definition 1.21: symmetric square root

For NND $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, $\mathbf{S}^{1/2} = \mathbf{U}\mathbf{\Lambda}^{1/2}\mathbf{U}^\top$; for PD $\mathbf{S}$, $\mathbf{S}^{-1/2} = (\mathbf{S}^{-1})^{1/2} = \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top$. $\mathbf{S}^{1/2}$ is symmetric with $\mathbf{S}^{1/2}\mathbf{S}^{1/2} = \mathbf{S}$.

5.3 Covariance matrices and ellipses

With $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ the ellipse $E_S = \{\mathbf{x} : \mathbf{x}^\top \mathbf{S}^{-1}\mathbf{x} = 1\}$ reads

$$\frac{(\mathbf{u}_1^\top \mathbf{x})^2}{\lambda_1} + \cdots + \frac{(\mathbf{u}_p^\top \mathbf{x})^2}{\lambda_p} = 1,$$

so its principal axes are along $\mathbf{u}_i$ with half-lengths $\sqrt{\lambda_i}$. The eigendecomposition of a covariance matrix gives (1) the rotation $\mathbf{x} \mapsto \mathbf{U}^\top \mathbf{x}$ to uncorrelated coordinates and (2) the sphering $\mathbf{x} \mapsto \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top \mathbf{x}$.

Theorem 1.22: covariance of rotated data (slides Discussion)

If $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, the rotated data $\mathbf{x}_i' = \mathbf{U}^\top \mathbf{x}_i$ have sample covariance $\mathbf{\Lambda}$.

[Supplement]. $\frac{1}{n-1}\sum_i \mathbf{U}^\top(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{U} = \mathbf{U}^\top \mathbf{S}\mathbf{U} = \mathbf{\Lambda}$. ■

The eigenvalues of a covariance matrix are the variances of mutually uncorrelated rotated coordinates — this is PCA.

6 Geometry of Principal Component Analysis

With many variables, inspecting all pairwise scatterplots is inefficient. PCA finds, in order, linear combinations of the original variables that explain the most variation, and summarizes the data by a few of them (details in Ch. 7).

6.1 Which direction to project on?

The scores $t_i(\mathbf{a}) = \mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}})$ along a unit vector $\mathbf{a}$ have sample variance $\frac{1}{n-1}\sum_i t_i(\mathbf{a})^2 = \mathbf{a}^\top \mathbf{S}\mathbf{a}$.

Definition 1.23: principal component directions

First direction $\mathbf{u}_1 \in \arg\max_{\|\mathbf{a}\|=1} \mathbf{a}^\top \mathbf{S}\mathbf{a}$; later directions maximize variance subject to orthogonality to the earlier ones. The solutions are eigenvectors of $\mathbf{S}$.

[Supplement] $\mathbf{u}_1$ is the top eigenvector. With $\mathbf{z} = \mathbf{U}^\top \mathbf{a}$ ($\|\mathbf{z}\| = 1$), $\mathbf{a}^\top \mathbf{S}\mathbf{a} = \sum_j \lambda_j z_j^2 \le \lambda_1$, with equality at $\mathbf{a} = \mathbf{u}_1$. Under $z_1 = 0$ the maximum is λ_2 at $\mathbf{u}_2$, and so on. ■

6.2 Principal component scores

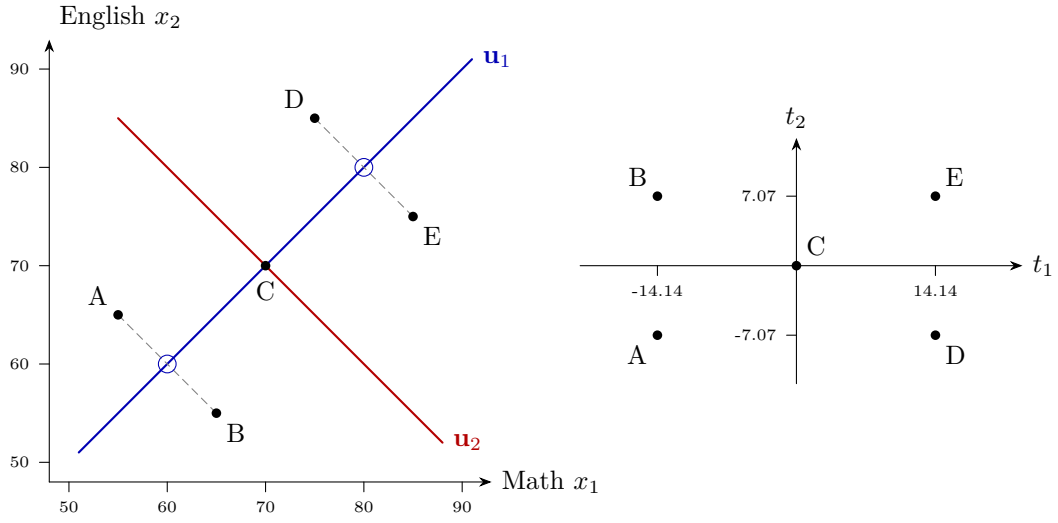
With $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, $\lambda_1 \ge \cdots \ge \lambda_p \ge 0$, the score matrix is $\mathbf{T} = \mathbf{X}_C\mathbf{U} = (t_{ij})$, $t_{ij} = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{u}_j$. Each column of $\mathbf{T}$ has mean 0 and $\frac{1}{n-1}\mathbf{T}^\top\mathbf{T} = \mathbf{U}^\top\mathbf{S}\mathbf{U} = \mathbf{\Lambda}$: the jth PC has sample variance λ_j and different PC scores are uncorrelated.

Example 1.24: scores in two subjects

Student	Math x_1	English x_2	PC1 score	PC2 score
A	55	65	-14.14	-7.07
B	65	55	-14.14	7.07
C	70	70	0	0
D	75	85	14.14	-7.07
E	85	75	14.14	7.07

$\bar{\mathbf{x}} = (70, 70)^\top$, $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$, $\lambda_1 = 200$, $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$, $\lambda_2 = 50$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$.

$t_{i1} = \{(x_{i1} - 70) + (x_{i2} - 70)\}/\sqrt{2}$ is the overall level; $t_{i2} = \{(x_{i1} - 70) - (x_{i2} - 70)\}/\sqrt{2}$ is the relative difference between the subjects. A and B share the same PC1 score, so PC1 alone cannot distinguish their relative strengths.



(a) original scores (b) principal component scores

Figure. *(a) Principal axes through $\bar{\mathbf{x}}$; A, B (and D, E) project to the same point on the first axis. (b) Score plot: PC2 separates A from B.*

6.3 Dimension reduction and reconstruction error

For student E, $\mathbf{x}_E = (85, 75)^\top$ and $t_{E1} = 10\sqrt{2}$. Reconstructing from PC1 gives $\hat{\mathbf{x}}_E = (70, 70)^\top + 10\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1,1)^\top = (80, 80)^\top$: the average level is kept, the difference between subjects is lost.

Theorem 1.25: reconstruction error

For the reconstruction $\hat{\mathbf{x}}_i = \bar{\mathbf{x}} + \sum_{j \le q} t_{ij}\mathbf{u}_j$ from the first q PCs,

$$\sum_{i=1}^{n} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = (n-1) \sum_{j=q+1}^{p} \lambda_j.$$

Among all q-dimensional planes through $\bar{\mathbf{x}}$, PCA minimizes the reconstruction error, which is equivalent to maximizing the total projected variance.

[Supplement]. $\mathbf{x}_i - \hat{\mathbf{x}}_i = \sum_{j>q} t_{ij}\mathbf{u}_j$, so $\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = \sum_{j>q} t_{ij}^2$ and $\sum_i t_{ij}^2 = (n-1)\lambda_j$. For any

orthonormal $\mathbf{V} \in \mathbb{R}^{p\times q}$, Pythagoras gives $\sum_i \|(\mathbf{x}_i-\bar{\mathbf{x}})-\mathbf{V}\mathbf{V}^\top(\mathbf{x}_i-\bar{\mathbf{x}})\|^2 = (n-1)\{\mathrm{tr}\mathbf{S}-\mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})\}$, so minimizing the error is maximizing $\mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})$, whose maximum $\lambda_1+\cdots+\lambda_q$ follows from von Neumann's inequality (Ch. 2) or Ch. 7. ■

6.4 Proportion of variance, scale and caveats

- Since $\mathrm{tr}\mathbf{S} = \sum_k \lambda_k$, the proportion of variance explained by PC j is $\lambda_j/\sum_k \lambda_k$ and the cumulative proportion is $\sum_{j\le q}\lambda_j/\sum_k\lambda_k$ ($80\% = 200/250$ in the example). Choose q from the cumulative proportion or a scree plot.
- Covariance-based PCA depends on units (rescaling English to 1000 points multiplies its variance by 100). PCA of standardized data $z_{ik} = (x_{ik}-\bar{x}_k)/s_k$ is **correlation PCA**. If differences in variance are meaningful, standardizing is inappropriate.
- High explained variance does not mean good group separation. With $Y \in \{-1,1\}$ equally likely, $X_1 = 10Z$, $X_2 = Y$, $Z \sim N(0,1) \perp\!\!\!\perp Y$: $\mathbf{\Sigma} = \mathrm{diag}(100,1)$ and PC1 explains $100/101 \approx 99\%$, but $X_1 \perp\!\!\!\perp Y$ while X_2 alone separates the groups perfectly. PCA keeps variation, not predictive information.
- Still, for the Golub data ($p = 7129$; slides Example 1.8), the PC1–PC2 plot, built without the labels, separates ALL/AML better than the original variables: PCA can be used to *explore* group differences.

7 Random Vectors and the Geometry of Variable Vectors

Motivation. Viewing each variable as a vector in $\mathbb{R}^n$ gives covariance and correlation a geometric meaning: lengths are standard deviations and angles are correlations. This picture makes projection arguments (regression, partial correlation) transparent.

7.1 Random vectors

The data $\mathbf{X} = [\mathbf{x}_1,\ldots,\mathbf{x}_n]^\top$ are a random sample from a population modeled by the joint distribution of $\mathbf{x} = (X_1,\ldots,X_p)^\top \sim f$; a simple random sample is i.i.d., $\mathbf{x}_1,\ldots,\mathbf{x}_n \overset{\text{iid}}{\sim} f$.

Example 1.26: distributions of random vectors (slides Example 2.1)

1. Independent $X_i \sim \mathrm{Unif}(a_i,b_i)$: $f(\mathbf{x}) = \prod_i \frac{1}{b_i-a_i}\mathbf{1}\{x_i \in (a_i,b_i)\}$, uniform on a hypercube.
2. Independent $X_i \sim N(0,1)$: $f(\mathbf{x}) = (2\pi)^{-p/2}e^{-\|\mathbf{x}\|^2/2}$, the standard multivariate normal.
3. (X_1,X_2) uniform on $0<x_1<x_2<1$: $f = 2\cdot\mathbf{1}\{0<x_1<x_2<1\}$. $X_1 \mid X_2 = x_2 \sim \mathrm{Unif}(0,x_2)$ depends on x_2, so they are not independent.
4. Bivariate normal with correlation ρ: $f = \frac{1}{2\pi\sqrt{1-\rho^2}}\exp\{-\frac{x_1^2-2\rho x_1x_2+x_2^2}{2(1-\rho^2)}\}$. Both marginals are $N(0,1)$ but the joint density is not their product: not independent.

Definition 1.27: moments of a random vector

$$\boldsymbol{\mu} = \mathrm{E}(\mathbf{X}), \qquad \mathrm{Cov}(\mathbf{X}) = \mathrm{E}\{(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top\} = \mathrm{E}(\mathbf{X}\mathbf{X}^\top)-\boldsymbol{\mu}\boldsymbol{\mu}^\top, \qquad \mathrm{Cov}(\mathbf{X},\mathbf{Y}) = \mathrm{E}\{(\mathbf{X}-\mathrm{E}\mathbf{X})(\mathbf{Y}-\mathrm{E}\mathbf{Y})^\top\}.$$

For $\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{X}) \succ 0$, $d_{\boldsymbol{\Sigma}}(\mathbf{x}, \boldsymbol{\mu})^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$. Generalized variance $|\mathrm{Cov}(\mathbf{X})|$, total variance $\mathrm{tr}\mathrm{Cov}(\mathbf{X})$; sample versions $|\mathbf{S}|$, $\mathrm{tr}\mathbf{S}$.

Theorem 1.28: characterization of covariance matrices

$\boldsymbol{\Sigma}$ is a covariance matrix iff $\boldsymbol{\Sigma}$ is symmetric and NND.

Proof. ($\Rightarrow$) [Supplement] $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \mathrm{Var}(\mathbf{a}^\top \mathbf{X}) \geq 0$.

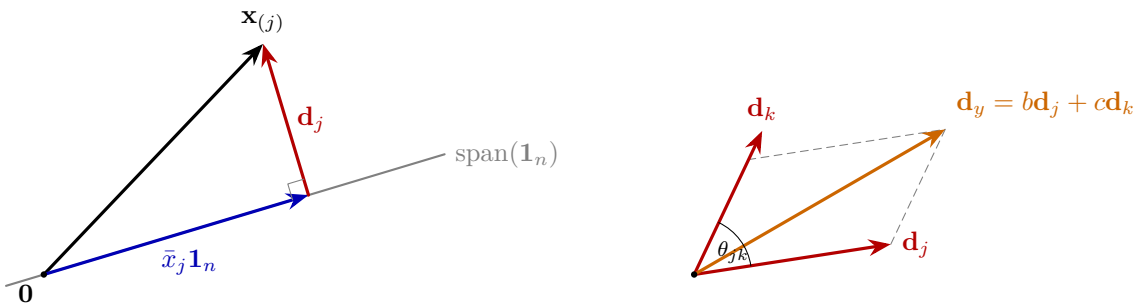
($\Leftarrow$) $\mathbf{X} = \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$, $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, has $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma}$. ■

7.2 Mean, variance and correlation through variable vectors

Viewing the variable vectors $\mathbf{x}_{(j)} \in \mathbb{R}^n$ (p vectors in n-space) gives a geometric reading of the statistics.

- **Mean**: $\bar{x}_j = \frac{1}{n}\mathbf{1}_n^\top \mathbf{x}_{(j)}$. The projection of $\mathbf{x}_{(j)}$ onto $\mathbf{1}_n$ is $P_{\mathbf{1}_n}(\mathbf{x}_{(j)}) = \mathbf{1}_n(\mathbf{1}_n^\top \mathbf{1}_n)^{-1}\mathbf{1}_n^\top \mathbf{x}_{(j)} = \bar{x}_j \mathbf{1}_n$ with score $\sqrt{n}\,\bar{x}_j$.
- **Deviation vector**: $\mathbf{d}_j = \mathbf{x}_{(j)} - \bar{x}_j \mathbf{1}_n$; $\mathbf{x}_{(j)} = \bar{x}_j \mathbf{1}_n + \mathbf{d}_j$ is an orthogonal decomposition ($\mathbf{1}_n^\top \mathbf{d}_j = 0$).
- $s_j = \|\mathbf{d}_j\|/\sqrt{n-1}$, $s_j^2 = \|\mathbf{d}_j\|^2/(n-1)$, $s_{jk} = \mathbf{d}_j^\top \mathbf{d}_k/(n-1)$.
- **The sample correlation is the cosine of the angle between deviation vectors**: $r_{jk} = \dfrac{\mathbf{d}_j^\top \mathbf{d}_k}{\|\mathbf{d}_j\|\|\mathbf{d}_k\|} = \cos\theta_{jk}$.

A longer deviation vector means more variability. $\mathbf{d}_j = c\mathbf{d}_k$ with $c > 0$ ($\theta = 0$) gives $r = 1$, with $c < 0$ ($\theta = \pi$) gives $r = -1$; orthogonal deviation vectors give zero *linear* correlation.



(a) $\mathbf{x}_{(j)} = \bar{x}_j \mathbf{1}_n + \mathbf{d}_j$ (b) $r_{jk} = \cos\theta_{jk}$, $s_j \propto \|\mathbf{d}_j\|$

Figure. *Variables as vectors in $\mathbb{R}^n$: (a) mean part and deviation vector; (b) lengths are standard deviations, angles give correlations.*

Theorem 1.29: unbiasedness of $\bar{\mathbf{X}}$ and $\mathbf{S}$ (slides Result 2.1)

If $\mathbf{X}_1, \ldots, \mathbf{X}_n$ is a random sample with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$,

$$\mathrm{E}(\bar{\mathbf{X}}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\bar{\mathbf{X}}) = \tfrac{1}{n}\boldsymbol{\Sigma}, \qquad \mathrm{E}(\mathbf{S}_n) = \tfrac{n-1}{n}\boldsymbol{\Sigma}, \qquad \mathrm{E}(\mathbf{S}) = \boldsymbol{\Sigma}.$$

Proof. $\mathrm{E}\bar{\mathbf{X}} = \frac{1}{n}\sum \mathrm{E}\mathbf{X}_j = \boldsymbol{\mu}$. $\mathrm{Cov}(\bar{\mathbf{X}}) = \frac{1}{n^2}\sum_j \sum_\ell \mathrm{E}\{(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top\}$, and terms with $j \neq \ell$ vanish by independence, giving $\frac{1}{n^2} \cdot n\boldsymbol{\Sigma}$.

[Supplement] Since $\sum_j (\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})^\top = \sum_j (\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_j - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top$, its expectation is $n\boldsymbol{\Sigma} - \boldsymbol{\Sigma} = (n-1)\boldsymbol{\Sigma}$. ■

7.3 Linear combinations of variables

For the new variable $z = a + \mathbf{b}^\top \mathbf{x} = a + \sum_j b_j x_j$ the variable vector is $\mathbf{z} = a\mathbf{1}_n + \sum_j b_j \mathbf{x}_{(j)}$, with

$$P_{\mathbf{1}_n}(\mathbf{z}) = \Big(a + \sum_j b_j \bar{x}_j\Big)\mathbf{1}_n, \qquad \mathbf{d}_z = \sum_j b_j \mathbf{d}_j = \mathbf{X}_C \mathbf{b}.$$

Theorem 1.30: sample variance and covariance of linear combinations

$$\widehat{\mathrm{Var}}(z) = \tfrac{1}{n-1}\mathbf{d}_z^\top \mathbf{d}_z = \mathbf{b}^\top \mathbf{S}\mathbf{b}, \qquad y = \mathbf{a}^\top \mathbf{x},\ z = \mathbf{b}^\top \mathbf{x} \ \Rightarrow\ \widehat{\mathrm{Cov}}(y,z) = \tfrac{1}{n-1}\mathbf{d}_y^\top \mathbf{d}_z = \mathbf{a}^\top \mathbf{S}\mathbf{b}.$$

For two variables, $y = b\,x_1 + c\,x_2$ has $\widehat{\mathrm{Var}}(y) = \{b^2\|\mathbf{d}_1\|^2 + c^2\|\mathbf{d}_2\|^2 + 2bc\|\mathbf{d}_1\|\|\mathbf{d}_2\|\cos\theta\}/(n-1)$.

8 Dimension and Rank of the Data

Motivation. If $n \le p$, the sample covariance matrix is singular and cannot be inverted. The rank result here explains when classical multivariate procedures break down.

- Linear independence: $\sum c_i \mathbf{x}_i = \mathbf{0} \Rightarrow$ all $c_i = 0$. The rank is the maximal number of linearly independent columns (= rows). $\mathrm{rank}(\mathbf{X}) \le \min(n, p)$.
- $\mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\mathbf{A}^\top) = \mathrm{rank}(\mathbf{A}^\top\mathbf{A})$, hence $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C^\top \mathbf{X}_C) = \mathrm{rank}(\mathbf{X}_C)$.
- All deviation vectors are orthogonal to $\mathbf{1}_n$, so they lie in an $(n-1)$-dimensional subspace: $\mathrm{rank}(\mathbf{X}_C) \le \min(n-1, p)$.
- An $m \times k$ matrix is **full-rank** if its rank is $\min(m,k)$ and **rank-deficient** otherwise. If $p > n$, $\mathrm{rank}(\mathbf{S}) \le n - 1 < p$: $\mathbf{S}$ is always rank-deficient.

Theorem 1.31: rank-deficiency property (slides Proposition, homework)

For the covariance matrix $\mathbf{S}$ of $\mathbf{X} \in \mathbb{R}^{n\times p}$ the following are equivalent.

1. $\mathbf{S}$ is rank-deficient.
2. There is $\mathbf{a} \neq \mathbf{0}$ with $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all i.
3. All projection scores coincide: $s_{\mathbf{a}}(\mathbf{x}_i) = s_{\mathbf{a}}(\mathbf{x}_j)$ for all i, j.
4. $\mathrm{span}\{\mathbf{d}_1, \ldots, \mathbf{d}_p\}$ has dimension less than p.
5. Some deviation vector is a linear combination of the others.
6. At least one eigenvalue of $\mathbf{S}$ is 0.

Proof. (1)⇔(2): $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C^\top) = r < p$ iff the columns $\{\mathbf{x}_i - \bar{\mathbf{x}}\}$ of $\mathbf{X}_C^\top$ span a space of dimension $r < p$, iff a nonzero $\mathbf{a}$ in its orthogonal complement exists.

(2)⇔(3): $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0 \iff \mathbf{a}^\top\mathbf{x}_i = \mathbf{a}^\top\bar{\mathbf{x}}$ for all i.

(1)⇔(4): $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C) = \dim \mathrm{span}\{\mathbf{d}_j\}$.

(4)⇔(5): dimension below p means $\{\mathbf{d}_j\}$ is linearly dependent.

(2)⇒(6): $\mathbf{Sa} = \frac{1}{n-1}\mathbf{X}_C^\top \mathbf{X}_C \mathbf{a} = \mathbf{0} = 0 \cdot \mathbf{a}$.

(6)⇒(2): $\mathbf{Su} = \mathbf{0}$ gives $\mathbf{u}^\top \mathbf{Su} = \frac{1}{n-1}\|\mathbf{X}_C \mathbf{u}\|^2 = 0$, so $\mathbf{X}_C \mathbf{u} = \mathbf{0}$. ■

If $\mathbf{S}$ is full rank, every direction carries independent variation. If it is rank-deficient, some direction carries no information at all (an exact linear relation among the variables), the sphering $\mathbf{x} \mapsto \mathbf{S}^{-1/2}\mathbf{x}$ is undefined, and statistical analysis becomes less reliable. In high dimensions ($p > n$) $\mathbf{S}$ is always rank-deficient: $\operatorname{rank}(\mathbf{S}) \le \min\{n-1, p\} < p$.

Remark 1.32: generalized variance as a squared volume [Supplement]

For $p = 2$, $|\mathbf{S}| = s_{11}s_{22} - s_{12}^2 = (\|\mathbf{d}_1\|\|\mathbf{d}_2\| \sin\theta_{12})^2/(n-1)^2$: the squared area of the parallelogram spanned by $\mathbf{d}_1, \mathbf{d}_2$. In general $|\mathbf{S}| = (n-1)^{-p}(\text{volume})^2$, which is 0 exactly in the rank-deficient case.



weak correlation: $|\mathbf{S}|$ large strong correlation: $|\mathbf{S}|$ small

Figure. [Supplement] *A small angle between* $\mathbf{d}_1$ *and* $\mathbf{d}_2$ *flattens the parallelogram and* $|\mathbf{S}|$ *tends to 0.*